

Hasan Bukhari

Software Engineering • Mobile Systems • Backend Systems

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EDUCATION

University of Southern Mississippi

B.S. in Computer Science; Minor in Biological Sciences

GPA: 4.00 • President's List • Academic Excellence Scholar

Beaconhouse College Program

Cambridge A Levels: 4 As in Biology, Chemistry, Mathematics, and Physics

Expected Dec 2028

Hattiesburg, MS

May 2024

Karachi, Pakistan

EXPERIENCE

Data Analyst Intern, Health, Safety & Environment

Engro Technical Services

- Engineered Python, FastAPI, and Power Query pipelines for risk scoring, scheduled reporting, and automated change monitoring across approximately 1,750 controlled engineering and HSE files.
- Built Power BI semantic models, DAX measures, and executive dashboards over 130,000+ incident, near-miss, inspection, audit, waiver, and corrective-action records.

Jul 2026 – Aug 2026

Karachi, Pakistan

Student Worker, Operations Technology & Event Support

Thad Cochran Center / Union Operations, University of Southern Mississippi

- Built SetItUp and UnionInventoryPro to digitize room requests, event layouts, equipment tracking, and recurring inventory audits for university operations.

Jun 2024 – Present

Hattiesburg, MS

Undergraduate Researcher, Ichthyology Lab

University of Southern Mississippi

- Build reproducible Python workflows and chromosome-correspondence matrices to compare chromosome-scale *Fundulus* genome assemblies and evaluate synteny and candidate fusion/fission patterns.

Spring 2026 – Present

Hattiesburg, MS

Software Engineering Intern

Shahid I.T. Consultants

- Automated recurring record-update workflows in Python, reducing manual processing by 30%; supported integration testing, deployment documentation, and client application maintenance.

Aug 2023 – Feb 2024

Karachi, Pakistan

SELECTED SOFTWARE PROJECTS

HEARTLINE — Native Android Lyric Companion | Kotlin, Jetpack Compose, Room, Android Services | [GitHub](#)

2026

- Built a native Android app that detects active media sessions, synchronizes lyrics with playback, caches tracks offline, renders live notifications, and creates customizable lyric images entirely on-device.
- Engineered resilient notification-listener recovery with bounded reconnect backoff and health checks; implemented Room persistence, secure FileProvider/MediaStore sharing, accessibility semantics, unit tests, Android lint, and GitHub Actions APK builds.

AEGIS — Controlled Document Intelligence & Governance | Python, FastAPI, React, TypeScript, Tauri, SQLite | [GitHub](#)

2026

- Built a local-first Windows platform for deterministic multi-format document monitoring, version retention, change evidence, governance workflows, and tamper-evident audit history.
- Engineered trusted-baseline reconciliation, SHA-256 integrity, persistent document identity, adversarial failure handling, and native Windows packaging; validated with 55 regression/adversarial tests and synthetic workloads up to 5,000 documents and 500,000 diff fragments.

Taaqat & Nivala — Offline-First Flutter Apps | Dart, Flutter, Material 3, Local Persistence | [GitHub](#)

2026

- Built two offline-first Android apps: Taaqat, a workout platform with 172 exercises and recovery-aware planning, and Nivala, a Pakistani nutrition tracker with 278 localized foods and recipes.
- Implemented versioned local persistence, migration support, validated import/export, native reminders, automated tests, static analysis, and CI pipelines producing APK/AAB artifacts.

BioLitGraph — Biomedical Literature Graph Explorer | Python, Flask, PubMed/PubTator, NetworkX | [GitHub](#)

2026

- Built a biomedical literature-mining application that normalizes extracted entities into relationship graphs while preserving paper-level evidence and exportable CSV/JSON tables.

To Ash Again — 2D Narrative Platformer | Python, Pygame, OOP | [GitHub](#)

2025

- Architected a four-act platformer with modular player, enemy AI, progression, inventory, save-state, cutscene, and multi-phase boss systems across a 258-file codebase.

TECHNICAL SKILLS

Languages: Python, C++, Kotlin, Dart, JavaScript, SQL

Mobile & Frontend: Android SDK, Jetpack Compose, Flutter, Material 3, React, TypeScript, HTML/CSS

Backend & Data: FastAPI, Flask, REST APIs, SQLAlchemy, Room, pandas, NumPy, NetworkX, Power BI, DAX, Power Query

Tools & Infrastructure: Git/GitHub, GitHub Actions, Linux/WSL, Docker, Tauri, PyInstaller, Gradle, CI/CD

LEADERSHIP & HONORS

President & Founder, Student Poets Association and Creative Writing Club, USM — Lead arts programming, public showcases, open mics, and club operations.

Selected Honors: Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026); President's List.